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CIRCULATORY SUPPORT DEVICES, SYSTEMS, AND METHODS

Abstract

A circulatory support system may include a blood pump and a removable controller configured to control operation of the blood pump and be removably coupled to a console. The removable controller comprising a display configured to display status indicators corresponding to parameters related to operation of the blood pump; speed controls configured to control a speed of the blood pump; a first connector configured to communicatively couple the removable controller with the blood pump; and a second connector configured to communicatively couple the removable controller with the console.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/559,362, filed Feb. 29, 2024, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure pertains to mechanical circulatory support devices, systems, and methods. More specifically, the present disclosure relates to percutaneous ventricular assist device (PVAD) systems with removable controllers.
BACKGROUND
[0003] A wide variety of intracorporeal and extracorporeal medical devices and systems have been developed for medical use, for example, in cardiac procedures and/or for cardiac treatments. Some of these devices and systems include guidewires, catheters, catheter systems, pump devices, circulatory assist devices, and the like. These devices and systems are manufactured by any one of a variety of different manufacturing methods and may be used according to any one of a variety of methods. Of the known medical devices, systems, and methods, each has certain advantages and disadvantages. There is an ongoing need to provide alternative medical devices and systems as well as alternative methods for manufacturing, display, operation and optimization of medical devices and systems.
BRIEF SUMMARY
[0004] This disclosure provides design, material, manufacturing method, and use alternatives for medical devices, including ventricular assist devices.
[0005] A first example is a circulatory support system. The circulatory support system includes a blood pump and a removable controller configured to control operation of the blood pump and be removably coupled to a console. The removable controller includes a display configured to display status indicators corresponding to parameters related to operation of the blood pump, speed controls configured to control a speed of the blood pump, a first connector configured to communicatively couple the removable controller with the blood pump, and a second connector configured to communicatively couple the removable controller with the console.
[0006] Alternatively or additionally to any of the examples herein, in another example, the first connector is disposed at a first surface of the removable controller, and the second connector is disposed at a second surface of the removable controller.
[0007] Alternatively or additionally to any of the examples herein, in another example, the first surface is a first side surface and the second surface is a second side surface that is opposite the first side surface.
[0008] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to be disposed at least partially within a docking port of the console.
[0009] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to be disposed entirely within the docking port of the console.
[0010] Alternatively or additionally to any of the examples herein, in another example, the first surface is coplanar with an outer surface of the console when the removable controller is disposed entirely within the docking port of the console.

[0011] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured such that at least the first connector remains physically accessible when the removable controller is removably coupled to the console.
[0012] Alternatively or additionally to any of the examples herein, in another example, at least the first connector and the display remain accessible when the removable controller is removably coupled to the console.
[0013] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to be in wired communication, wireless communication, or both, with the console.
[0014] Alternatively or additionally to any of the examples herein, in another example, the removable controller includes a rechargeable battery.
[0015] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to receive wireless power, wired power, or both, to recharge the rechargeable battery.
[0016] Alternatively or additionally to any of the examples herein, in another example, the system includes a wearable device configured to receive the removable controller. The removable controller is configured to be removably coupled to the wearable device.
[0017] Alternatively or additionally to any of the examples herein, in another example, the removable controller includes a third connector disposed on an outer surface of the removable controller. The third connector is configured to removably couple the removable controller to the wearable device.
[0018] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to remain communicatively coupled to the blood pump when the removable controller is removably coupled to the console to operate the blood pump.
[0019] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to remain communicatively coupled to the blood pump when the removable controller is decoupled from the console to operate the blood pump.
[0020] Another example is a circulatory support system. The circulatory support system includes a blood pump, a console, and a removable controller. The console includes a display configured to display at least a first set of status indicators corresponding to parameters related to operation of the blood pump, a docking port, and a connector. The removable controller includes a user interface configured to display a second set of status indicators corresponding to parameters related to operation of the blood pump, speed controls configured to control a speed of the blood pump, a first connector configured to communicatively couple the removable controller with the blood pump, and a second connector configured to connect to the connector of the console when the removable controller is docked with the console via the docking port to communicatively couple the removable controller to the console.
[0021] Alternatively or additionally to any of the examples herein, in another example, the docking port is configured to obscure at least the display of the removable controller when the removable controller docked with the console via the docking port.
[0022] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to cease display of the second set of status indicators when communicatively coupled to the console.
[0023] Alternatively or additionally to any of the examples herein, in another example, the display of the removable controller remains exposed when the removable controller is docked with the console via the docking port and the removable controller is configured to continue to display the second set of status indicators when communicatively coupled to the console.
[0024] Another example is a circulatory support system. The circulatory support system includes a blood pump, a console, and a removable controller. The console includes a display configured to display at least a first set of status indicators corresponding to parameters related to operation of the

blood pump, a docking port, a connector, and a first battery. The removable controller includes a display configured to display a second set of status indicators corresponding to parameters related to operation of the blood pump, speed controls configured to control a speed of the blood pump, a first connector configured to communicatively couple the removable controller with the blood pump, a second connector configured to connect to the connector of the console when the removable controller is docked with the console via the docking port to communicatively couple the removable controller with the console, and a second battery.
[0025] Alternatively or additionally to any of the examples herein, in another example, the removable controller is configured to maintain operation of the blood pump when undocked from the console.
[0026] The above summary of some embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify some of these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0027] The disclosure may be more completely understood in consideration of the following detailed description in connection with the accompanying drawings, in which:
[0028] FIG. **1**A is a schematic diagram of an illustrative circulatory support system with a blood pump positioned in a patient;
[0029] FIG. **1**B is a schematic diagram of another illustrative circulatory support system with a blood pump positioned in a patient;
[0030] FIG. **2** is a schematic diagram of an illustrative computing system including a removable controller;
[0031] FIG. **3** is a schematic diagram of a portion of an illustrative circulatory support system with a removable controller coupled to a console;
[0032] FIG. **4** is a schematic diagram of a portion of an illustrative circulatory support system with the removable controller decoupled from the console;
[0033] FIG. **5** is a schematic diagram showing a plurality of surfaces of the removable controller;
[0034] FIG. **6** is a schematic diagram showing a removable controller coupled to a patient; and
[0035] FIG. **7** is a schematic diagram of a portion of another illustrative circulatory support system with a removable controller coupled to a console.
[0036] While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.
DETAILED DESCRIPTION
[0037] For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.
[0038] All numeric values are herein assumed to be modified by the term
[0039] “about,” whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (i.e., having the same function or result). In many instances, the term “about” may include numbers that are rounded to the nearest significant figure.
[0040] The recitation of numerical ranges by endpoints includes all numbers within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).
[0041] As used in this specification and the appended claims, the singular forms “a”, “an”, and

"the" include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term "or" is generally employed in its sense including "and/or" unless the content clearly dictates otherwise.

[0042] It is noted that references in the specification to "a configuration", "another configuration", "some configurations", "other configurations", etc., indicate that the configuration described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one configuration, it should be understood that such features, structures, and/or characteristics may also be used in connection with other configurations whether or not explicitly described unless clearly stated to the contrary.

[0043] The following detailed description should be read with reference to the drawings in which similar structures in different drawings are numbered the same. The drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the disclosure. Additionally, it should be noted that in any given figure, some features may not be shown, or may be shown schematically, for clarity and/or simplicity. Additional details regarding some components and/or method steps may be illustrated in other figures in greater detail. The devices and/or methods disclosed herein may provide a number of desirable features and benefits as described in more detail below.

[0044] A variety of circulatory support devices and systems are known for assisting or replacing a pumping function of a heart in a patient with severe heart failure and/or other cardiac conditions. Circulatory support devices may be configured to treat patients with cardiogenic shock, myocardial infarction, acutely decompensated heart failure, and/or other heart related conditions. Additionally or alternatively circulatory support devices may support a patient during percutaneous coronary interventions and/or other procedures.

[0045] Example cardiac circulatory support devices include, but are not limited to, ventricular assist devices (VADs), total artificial hearts, intra-aortic balloon pumps (IABP), and extracorporeal membrane oxygenation (ECMO). Example VADs include left ventricular assist devices (LVADs), right ventricular assist devices (RVADs), and biventricular assist devices (BiVADs). A further illustrative VAD is a percutaneous ventricular assist device (PVAD), which may be inserted into a ventricle (e.g., a left ventricle or a right ventricle) of a heart of a patient via delivery through a femoral artery or vein and/or other suitable vasculature to the ventricle. As a PVAD may be placed at a desired location of anatomy of a patient via percutaneous access and delivery, the PVAD may be used in emergency medicine, a catheter laboratory, and/or other surgical and/or non-surgical settings.

[0046] FIG. **1**A depicts an illustrative percutaneous circulatory support system **10** including a circulatory support device **12** positioned in the heart **14** of a patient **16**. The circulatory support device **12** may include, among other components, a flexible elongate catheter shaft **20** having a first end attached to a handle **22** and a second end attached to a blood pump **24**. As depicted in FIG. **1**A, the blood pump **24** may be positioned in the left ventricle **18** of the patient **16**. The blood pump **24** may be delivered (e.g., tracked) to the left ventricle **18** percutaneously over a guidewire. For example, the catheter shaft **20** and the blood pump **24** may be tracked over a guidewire through the femoral artery and the descending aorta, over the aortic arch, through the ascending aorta, past the aortic valve, and into the left ventricle **18**.

[0047] In some examples, the blood pump **24** may be positioned within the heart **14** such that a blood inlet positioned along the distal region of the blood pump **24** may be located in the left ventricle **18** and a blood outlet positioned along a housing of the blood pump **24** may be located in the aorta. Additionally, the blood pump **24** may include an electrically powered motor that drives rotation of an impeller (e.g., where the motor and/or the impeller may be positioned within the housing of the blood pump **24**). In some examples, the motor may power the rotation of the impeller via electromagnetic induction, but other suitable configurations are contemplated. The rotating impeller may draw blood from the left ventricle (via the blood inlet) into the aorta (via the blood outlet). In other words, an electrically powered motor may drive the impeller to pump blood from the left ventricle through the aortic valve and into the ascending aorta.

[0048] FIG. **1**A further illustrates that a first end of the catheter shaft **20** may be attached to a housing **22**. The housing **22** may be considered a connector and include a distal end region attached to the catheter shaft **20** and a proximal end region attached to a cable **26**. It can be appreciated that the housing **22** may include one or more actuators (e.g., buttons, levers, dials, switches, etc.) designed to permit a clinician to control various functions of the blood pump **24**. For example, a clinician may be able to control the speed of the motor and/or the impeller located in the blood pump **24** via actuation of an actuator located on the housing **22**, in some instances.

[0049] The cable **26** may include a proximal end region directly connected to a removable controller **36**, as described herein. That is, in contrast to other approaches such as those that may directly couple the cable **26** to the console **28**, the approaches herein directly couple the cable **26** to the removable controller **36** which can, in turn, be removably coupled to the console **28** (e.g., can be removably disposed within a docking port of the console **28**, removably attached to the top or side of the console **28**, etc.). In removably coupling the removable controller **36** to the console **28**, electrical connections between the removable controller **36** and the console **28** may be established, providing electrical communication between the console **28** and the blood pump **24**, via the cable **26**. This indirect coupling of the cable **26** to the console **28** can promote aspects herein, such as permitting the removable controller **36** to be readily coupled to or decoupled from the console **28** (e.g., mounted to/in the console **28**, yet detached from the console **28**), and yet retain the communicative coupling between the removable controller **36** and the blood pump **24** which may permit ease of transport of a patient and yet permit uninterrupted operation, monitoring and/or control of the blood pump **24**. The cable **26** may include or be an electrical power cable configured to power the blood pump **24**, a communication cable configured to communicate data between the blood pump **24** and the console **28**, an electrical cable, an optical cable, and/or other suitable type of cable.

[0050] The percutaneous circulatory support system **10** may include a sensor positioned within and/or along the console **28**, the handle **22**, the catheter shaft **20**, and/or the blood pump **24**, where the sensor may be coupled with the console **28** and/or other data monitoring component via wired connection (e.g., an electrical and/or optical connection of the cable **26**) and/or a wireless connection. The sensor positioned within and/or along the console **28**, within the handle **22**, the catheter shaft **20**, and/or the blood pump **24** may be configured to sense a parameter of or related to operation of the circulatory support system **10**, operation of the circulatory support device **12**, and/or the patient **16**. The sensor positioned within and/or along the console **28**, the handle **22**, along the catheter shaft **20**, and/or the blood pump **24** may be designed or configured to sense and/or monitor any suitable parameters including, but not limited to, blood pressures (e.g., arterial pressure, venous pressure), blood velocity, blood flowrate, mean arterial pressure, impeller speeds, motor speeds, an electrical current provided to the motor, voltage provided to the motor, back-EMF from the motor, and/or other suitable parameters, along with any suitable combination or temporal pattern of signals corresponding to the parameters. Further, additional parameters (e.g., flow through or across the blood pump **24**) may be derived by processing combinations of sensed data in a time dependent manner.

[0051] As depicted in FIG. **1**A, the console **28** may include controls (e.g., buttons, knobs, dials, touch-sensitive areas, etc.) **30** and/or one or more displays **31**. FIG. **1**A depicts the console **28** with a first display **32** and a second display **34**, but other suitable configurations of the console **28** with a single display and/or with more than two displays are contemplated. An example circulatory support system including a console with two displays is disclosed in Appl. No. 63/457,935, filed on Apr. 7, 2023, and titled CIRCULATORY SUPPORT DEVICE SYSTEM, which is hereby

incorporated by reference in its entirety for any and all purposes.
[0052] Additionally, while FIG. **1**A illustrates the first display **32** and the second display **34** as being integrated into the console **28**, it is contemplated that the circulatory support system **10** may be designed such that the first display **32**, the second display **34**, or both of the first display **32** and the second display **34** are separate, distinct components of the circulatory support system **10**. In other words, the first display **32**, the second display **34**, or both of the first display **32** and the second display **34** may be separate stand-alone displays, apart from the console **28** (e.g., may instead be manifested as the display **52** in the removable controller when the removable controller **36** is removably coupled to the console **28**, as described herein). In some examples, the first display **32** and the second display **34** may receive data from the same or different sources.
[0053] As detailed herein, in some instances the first display **32**, the second display **34**, or both of the first display **32** and the second display **34** may be separate components apart from the console **28** that are included in a removable controller **36**. The removable controller **36** can be analogous to the computing device or removable controller **36** described herein with respect to FIG. **2**.
[0054] As detailed herein, the removable controller **36** can be configured to control operation of the blood pump **24** in the absence of the console **28**, and can be removably coupled to a console **28**. For instance, the removable controller **36** can be configured to control operation of the blood pump **24** alone (e.g., when decoupled from the console **28**) and may thereby promote aspects herein promoting ease of transporting a patient (in which the blood pump **24** is disposed) via an emergency medical services (EMS) (e.g., via helicopter, ambulance, etc.) and/or transporting the patient between locations within a medical facility, such as a hospital, (e.g., between a catheter laboratory, an operating room, an intensive care unit, and/or other location within the medical facility), and yet permit control and monitoring of the operation of the blood pump **24**. For example, the removable controller **36** may be configured to couple to a belt, harness, sling, etc., and/or otherwise be coupled to the patient and thereby promote transportation and/or location of the patient, as compared to other approaches which require the presence of a bulky console (e.g., a console including or coupled to an elongate stand **13** (e.g., on top of which the console is disposed) and/or the console **28** including an external power supply **17**, etc.).
[0055] For instance, FIG. **1**B is a schematic diagram of another illustrative percutaneous circulatory support system **11** with a blood pump positioned in a patient. The percutaneous circulatory support system **11** is analogous to the percutaneous circulatory support system **10** in FIG. **1**A, but with the change that the removable controller **36** has been removed or decoupled from the console **28** and the console **28** is no longer present (e.g., is no longer coupled to the patient). For instance, in some embodiments the removable controller **36** may include a battery (e.g., a rechargeable battery), which as detailed herein, may promote ease of transporting of the patient in the absence of requiring the patient to remain coupled to a power supply or other external power circuitry configured to power the bulky console **28**. That is, the console **28** may be removably coupled to a removable controller **36**, as described herein. The removable controller **36** can include a user interface such as a display that is separate from a user interface retained with the console **28** and which is permanently located in the console **28**. For instance, FIG. **2** depicts a schematic box diagram of an illustrative computing device or removable controller **36** and console **28**. The computing device or removable controller **36** and/or the console **28** may be entirely or partially housed in a housing **40**. For instance, the console **28** may be permanently housed in the housing **40** and the computing device or removable controller **36** may be removably housed in the housing **40** and/or mounted on the housing **40**. The housing **40** may be an optional component, as represented by the broken lines defining the housing **40** depicted in FIG. **2**. Although various components are depicted as being included in the computing device or removable controller **36** and the console **28**, one more of the depicted components may be omitted and/or one or more additional or alternative components may be utilized.
[0056] The computing device or removable controller **36** may be any suitable computing device

configured to process data of or for the percutaneous circulatory support systems **10**, **11** and may be configured to facilitate operation of the percutaneous circulatory support systems **10**, **11**. The computing device or removable controller **36**, in some cases, may be configured to control operation of the blood pump **24** by establishing and/or outputting control signals to the blood pump **24** and/or components of or in communication with the console **28** to control and/or monitor operation of the blood pump **24**. In some examples, the removable controller **36** may be removably coupled to the console **28** and communicate with the blood pump **24** over a wired and/or wireless connection, but other suitable configurations are contemplated.

[0057] In some cases, the removable controller **36** and/or the console **28** may communicate with a remote server or other suitable computing device. The entire removable controller **36**, or at least a part of the removable controller **36**, is a component separate from a structure of the console **28**, that may communicate with electronic components of the blood pump **24**, the console **28**, and/or other suitable components of the percutaneous circulatory support systems **10**, **11** over one or more wired or wireless connections or networks (e.g., LANs and/or WANs).

[0058] The removable controller **36** may include, among other suitable components, a processor **42** (e.g., a hardware processor), a memory **44** (e.g., a medium storing non-transitory instructions thereon), an I/O unit **46**, a display **52**, and a battery **53**. Example other suitable components of the removable controller **36** that are not specifically depicted in FIG. **2** may include, but are not limited to, communication components, a touch screen, selectable buttons, wireless and/or wired communications components, power transmitting/and/or receiving components, and/or other suitable components of a removable controller. For instance, the removable controller **36** can include communications components such that the removable controller **36** is configured to be in wired communication, wired communication, or both, with the console **28**. For example, the removable controller **36** can be in wired and/or wireless communication with the console **28** when the removable controller **36** is disposed in the docking port **37** (see FIG. **4**). In other words, an electrical connection may be established between the removable controller **36** and the console **28** when the removable controller **36** is coupled to (e.g., docked with) the console **28**, and the electrical connection may be broken between the removable controller **36** and the console **28** when the removable controller **36** is decoupled from (e.g., undocked with) the console **28**.

[0059] In some instances, the removable controller **36** can be in wireless communication with the console **28** when the removable controller **36** is removed from the docking port **37** and remains proximate to the console **28** (e.g., is located on a common wireless network such as a common WiFi network as the console **28**). In such instances, the patient may be able to periodically walk or otherwise be moved a distance away from the console **28** and yet remain in communication with the console **28**. In an event in which the wireless communication with the console **28** is lost, the removable controller **36** may continue to operate the blood pump **24** uninterruptedly.

[0060] In some embodiments the removable controller **36** includes a battery **53** (e.g., a first battery). The battery **53** may be a rechargeable battery or a non-rechargeable battery. In some embodiments, the battery **53** is a recharge battery that is configured to be recharged in a wired and/or wireless manner. Stated differently, in some embodiments, the removable controller **36** can be configured to receive wireless power, wired power, or both, to charge the battery **53**. For instance, the battery **53** can be a rechargeable battery that is configured to be recharged in a wired manner when the removable controller **36** is coupled to (e.g., docked with) the console **28** (e.g., is disposed in the docking port of the console **28**) and/or a charger that is separate from the console **28**. For instance, the battery **53** may receive wired power via a (e.g., the second connector **66** as illustrated in FIG. **5**) and a corresponding connector (not illustrated) that is located in the docking port **37** of the console **28**.

[0061] In some examples, the console **28** can include a battery **51** (e.g., a second battery) separate from the battery **53** of the removable controller **36**. For instance, the battery **51** may be rechargeable battery or a non-rechargeable battery configured to supply power to the console **28** in the event of an unexpected loss of power to the console **28**.

[0062] The processor **42** of the removable controller **36** may include a single processor or more than one processor working individually or with one another. The processor **42** may be configured to receive and execute instructions, including instructions that may be loaded into the memory **44** and/or other suitable memory. Example components of the processor **42** may include, but are not limited to, central processing units, microprocessors, microcontrollers, multi-core processors, graphical processing units, digital signal processors, application specific integrated circuits (ASICs), artificial intelligence accelerators, field programmable gate arrays (FPGAs), discrete circuitry, and/or other suitable types of data processing devices.

[0063] The memory **44** of the removable controller **36** may include a single memory component or more than one memory component each working individually or with one another. Example types of memory **44** may include random access memory (RAM), EEPROM, flash, suitable volatile storage devices, suitable non-volatile storage devices, persistent memory (e.g., read only memory (ROM), hard drive, flash memory, optical disc memory, and/or other suitable persistent memory) and/or other suitable types of memory. The memory **44** may be or may include a non-transitory computer readable medium. The memory **44** may include instructions stored in a transitory state and/or a non-transitory state on a computer readable medium that may be executable by the processor **42** to cause the processor **42** to perform one or more of the methods and/or techniques described herein. Further, in some cases, the memory **44** and/or other suitable memory may store data received from the blood pump **24**.

[0064] The I/O unit **46** of the removable controller **36** may include a single I/O component or more than one I/O component each working individually or with one another. Example I/O unit **46** may be or may include any suitable types of communication hardware and/or software including, but not limited to, communication components or ports configured to communicate with electronic components of the percutaneous circulatory support systems **10**, **11**, and/or with other suitable computing devices or systems. Example types of I/O unit **46** may include, but are not limited to, wired communication components (e.g., HDMI components, Ethernet components, VGA components, serial communication components, parallel communication components, component video ports, S-video components, composite audio/video components, DVI components, USB components, optical communication components, and/or other suitable wired communication components), wireless communication components (e.g., radio frequency (RF) components, Low-Energy BLUETOOTH protocol components, BLUETOOH protocol components, Near-Field Communication (NFC) protocol components, WI-FI protocol components, optical communication components, ZIGBEE protocol components, and/or other suitable wireless communication components), and/or other suitable I/O units **46**.

[0065] The console **28** may be configured to communicate with the computing device or removable controller **36** via a wired and/or a wireless connection. The console **28** may include a display **31** (e.g., such as the first display **32** and/or the second display **34**, illustrated in FIG. **1**A), an input device **48**, an output device **49**, and/or one or more other suitable features.

[0066] The display **31** may be any suitable display. Example suitable displays include, but are not limited to, touch screen displays, non-touch screen displays, liquid crystal display (LCD) screens, light emitting diode (LED) displays, head mounted displays, virtual reality displays, augmented reality displays, and/or other suitable display types.

[0067] The input device(s) **48** may be and/or may include any suitable components and/or features for receiving user input via the console **28**. Example input device(s) **48** may include, but are not limited to, touch screens, keypads, mice, touch pads, microphones, selectable buttons, selectable knobs, optical inputs, cameras, gesture sensors, eye trackers, voice recognition controls (e.g., microphones coupled to appropriate natural language processing components) and/or other suitable input devices. In one example, the input devices **48** may include a touch screen that allows for setting set points and/or selecting selectable elements for additional detail concerning data or information associated with the selectable elements, but this is not required.

[0068] The output device(s) **49** may be and/or may include any suitable components and/or features for providing information and/or data to users and/or other computing components. Example output device(s) **49** include, but are not limited to, displays, speakers, vibration systems, tactile feedback systems, optical outputs, and/or other suitable output devices. In some instances, the output device(s) **49** and the input device(s) may be the same device such as a display.

[0069] The removable controller **36** of the percutaneous circulatory support system **10** may be configured to facilitate control of the blood pump **24** and store and/or monitor data related to operation of the blood pump **24** (e.g., data from operation of the blood pump **24**, data from control of the blood pump **24**, data from sensors of and/or associated with the blood pump **24**, and/or data from other suitable sources). The removable controller **36** may receive data over time from the blood pump **24** operating within a patient, where the received data may include patient data and/or data related to operation of the blood pump within the patient.

[0070] The removable controller **36** may be configured (e.g., sized and shaped) to be coupled to the blood pump **24** in a manner that allows for transporting the removable controller **36** (e.g., in the absence of the console **28**) with the patient in which the blood pump **24** has been positioned and currently operating, which may include transporting the patient via an emergency medical services (EMS) (e.g., via helicopter, ambulance, etc.) and/or transporting the patient between locations within a medical facility, such as a hospital, (e.g., between a catheter laboratory, an operating room, an intensive care unit, and/or other location within the medical facility). To assess the patient and/or operation of the blood pump **24**, medical professionals may need to evaluate historical patient data and/or blood pump **24** data gathered and/or saved at or by the removable controller **36**.

[0071] To facilitate transportation of the removable controller **36** with the patient, the removable controller **36** may be lightweight and/or compact. However, it may be desirable for the console **28** to have a large display configured to facilitate displaying patient data and/or blood pump data that facilitates medical professionals assessing the patient. That is, the system and approaches herein yield both a lightweight and/or compact removable controller **36** that is a readily transportable (e.g., when removed from the console **28**) to continue to operate the blood pump **24**, and yet provide the console **28** with a large display configured to facilitate displaying patient data and/or blood pump data that facilitates medical professionals readily assessing the patient once transported to a particular location (e.g., a particular room in a hospital).

[0072] Additionally, in some instances, the console **28** may include a first display, which may be configured to display a screen providing limited data or information (e.g., preconfigured screens with fixed layouts), and the removable controller **36** may include a second display (e.g., the display **52**) which may display screens having repetitive, condensed, or augmented views relative to what is depicted on the first display of the console **28**. Such a configuration of the displays and screens for the displays may allow for reducing a cost and/or size of the console **28** relative to when two integral (permanent) displays are required on the console **28** and/or when a large single display is included on the console **28**. An example of such a configuration is described in greater detail with respect to FIG. **7**. Although the two-display configuration is discussed herein with respect to percutaneous circulatory support systems, similar concepts may be applied to other medical systems that include a computing device with limited display space, such as medical systems configured to be transported with a patient, medical systems for use in operating rooms with limited space, and/or other suitable medical systems.

[0073] FIG. **3** is a schematic diagram of a portion of an illustrative circulatory support system **21** with a removable controller **36** coupled to (i.e., docked with) a console **28**, while FIG. **4** is a schematic diagram of a portion of an illustrative circulatory support system **21** with the removable controller **36** decoupled from (i.e., undocked from) the console **28**. As described herein, a blood pump (not illustrated in FIGS. **3-4**) may be coupled to, and remain coupled to, the removable controller **36** (e.g., electrically coupled via the cable **26**) when the blood pump is coupled to and/or is decoupled from the console **28**. As such, the blood pump and the removable controller **36** may facilitate transportation and/or locomotion of a patient, and yet the blood pump may remain communicatively coupled to the removable controller **36** such that the blood pump may remain in uninterrupted operation. For instance, the removable controller **36** is configured at least due to the location of the first connector **56** and/or the second connector (e.g., the second connector **66** as illustrated in FIG. **5**) and/or the inclusion of a battery or another portable power source in the removable controller **36** that is separate from a power source in the console **28** to remain communicatively coupled to the blood pump when the removable controller **36** is removably coupled to and/or is decoupled from the console **28**. Accordingly, the blood pump may remain communicatively coupled to the removable controller **36** when the removable controller **36** is decoupled from or is coupled to the console **28**, thereby permitting uninterrupted monitoring and/or control over operation of the blood pump, in contrast to other approaches that may, at least temporarily, disconnect a blood pump from a controller prior to, during, and/or subsequent to transportation (e.g., transportation in an ambulance) and/or locomotion of a patient (e.g., when a patient wishes to walk around or outside of a hospital room or other room in which they are primarily staying).

[0074] The console **28** and the removable controller **36** are analogous to the console **28** and the removable controller **36** discussed herein (e.g., discussed previously with respect to FIGS. **1**A, **1**B, and **2**). For instance, the console **28** can include controls **30**, a first display **32** and a second display **34**, as illustrated in FIG. **3**.

[0075] The console **28** can include a housing **40**. The housing **40** can include a plurality of surfaces such as a first surface **27-1**, a second surface **27-2**, a third surface **27-3**, a fourth surface **27-4**, a fifth surface **27-5**, and a sixth surface **27-6**. The first surface **27-1** can be a top surface of the housing **40** and the second surface **27-2** can be a bottom surface of the housing **40** that is opposite the first surface **27-1**. The third surface **27-3** can be a first side surface of the housing **40** and the fourth surface **27-4** can be a second side surface of the housing **40** that is opposite the third surface **27-3**. The fifth surface **27-5** can be a front surface of the housing **40** and a sixth surface **27-6** can be a back surface of the housing **40** that is opposite fifth surface **27-5**.

[0076] Similarly, the removable controller **36** can include a housing **33**. The housing **33** can be separable from the housing **40** of the console **28**. The housing **33** can include a plurality of surfaces such as a first surface **39-1**, a second surface **39-2**, a third surface **39-3**, a fourth surface **39-4**, a fifth surface **39-5**, and a sixth surface **39-6**. The first surface **39-1** can be a top surface of the housing **33** and the second surface **39-2** can be a bottom surface of the housing **33** that is opposite the first surface **39-1**. The third surface **39-3** can be a first side surface of the housing **33** and the fourth surface **39-4** can be a second side surface of the housing **33** that is opposite the third surface **39-3**. The fifth surface **39-5** can be a front surface of the housing **33** and a sixth surface **39-6** can be a back surface of the housing **33** that is opposite fifth surface **39-5**.

[0077] As illustrated in FIG. **3**, the console **28** can include the controls **30**, the first display **32** and the second display **34** disposed on the fifth surface **27-5** (e.g., front surface) of the housing **40**. However, other configurations where the controls **30**, the first display **32** and/or the second display **34** are disposed on a different surface of the housing **40** and/or a surface of the housing **33** are possible. For instance, as described herein in some instances the first display **32**, the second display **34**, or both the first display **32** and the second display **34** can be replaced by a display on a surface of the removable controller **36**. Similarly, in some examples the control **30** on the console **28** may be replaced by controls located on a surface of the removable controller **36**. Stated differently, in some embodiments, the console **28** may not include controls **30** and may instead be configured to removably coupled to the removable controller **36** which includes controls (e.g., the controls **50** such as speed controls and/or speed lock controls, etc.).

[0078] As illustrated in FIG. **3**, the console **28** can include a docking port **37**. The docking port **37** refers to a cavity formed within a surface of the console **28** or a receptacle formed on a surface of the console **28**. For instance, as illustrated in FIG. **3**, the docking port **37** can be formed in a side surface such as the first surface **27-1** of the console **28**.

[0079] In some embodiments, the removable controller **36** and the docking port **37** can be configured (e.g., sized and shaped) such that the removable controller **36** is disposed at least partially within the cavity of the docking port **37** of the console **28**. For instance, the removable controller **36** can have substantially the same shape as the cavity of the docking port **37**, as illustrated in FIG. **4**. For instance, the docking port **37** can be a substantially rectangular shaped docking port configured to receive a substantially rectangular shaped removable controller **36**, as illustrated in FIGS. **3-4**. Having the removable controller **36** be disposed at least partially within the docking port **37** can promote aspects herein such as securely and removably coupling the removable controller **36** to the docking port **37**. For instance, as illustrated in FIG. **3**, the removable controller **36** can be configured such that the removable controller **36** is disposed entirely within the docking port **37** of the console **28** when the removable controller **36** is removably coupled to the console **28**.

[0080] The docking port **37** can include a corresponding connector (not shown) that is configured to couple to a connector (e.g., a second connector) in the removable controller **36**. For instance, the docking port **37** may be a substantially rectangular shaped docking port having a corresponding connector configured couple to a connector (e.g., the second connector **66** on the second surface **39-2**, as illustrated in FIG. **5**) in the removable controller **36**. That is, the corresponding connector in the docking port **37** can be coupled to a connector (e.g., the second connector **66**, as illustrated in FIG. **5**) to communicatively couple the removable controller **36** with the console **28**. The removable controller **36** and the console **28** may exchange information and/or power when communicatively coupled. For instance, when communicatively coupled the removable controller **36** and/or the console **28** may be configured to control and/or otherwise communicate information with the blood pump and/or the console **28**.

[0081] In some instances, the docking port **37** can extend from an outer surface of the console **28** into a cavity of the console **28**. For instance, the docking port **37** can extend into the first surface **27-1** and be located proximate to both the fourth surface **27-4** (e.g., the bottom surface) and the sixth surface **27-6** (e.g., the back surface **27-6**), as illustrated in FIGS. **3-4**. Having the docking port **37** located in a side surface and be located proximate to both the fourth surface **27-4** (e.g., the bottom surface) and the sixth surface **27-6** (e.g., the back surface), as illustrated in FIGS. **3-4**, can promote aspects herein such as protecting a display **52** of the removable controller **36** when the removable controller **36** is disposed in the docking port **37**, and permitting access to the first connector **56** when the removable controller **36** is disposed in the docking port **37**. For instance, the docking port **37** may be configured to obscure (e.g., protect) at least the display **52** of the removable controller **36** when the removable controller **36** is disposed in the docking port **37**. For example, as illustrated in FIG. **3**, the docking port **37** may be configured to obscure both the controls (e.g., controls **50** as illustrated in FIG. **5**) and the display **52** and thereby protect the controls and the display **52** from unintended contact. In such embodiments, the removable controller **36** may be configured to cease operation (e.g., cease registering inputs from and/or cease/reduce power provided to the speed controls and/or the display **52**. Ceasing operation of the controls and/or the display **52** may reduce power consumption of the removable controller **36** and/or may mitigate any unintended input received via the controls and/or the display **52** while the removable controller **36** is disposed in the docking port **37**, and thereby may enhance operation of the percutaneous circulatory support system **10**. For instance, in some embodiments, the removable controller **36** can be configured to cease display of status indicators when the removable controller **36** is removably coupled to the docking port **37** of the console **28**.

[0082] However, the docking port **37** may be located in/on and/or proximate to another surface of the console. For instance, the docking port **37** may be located in the first surface **27-1** of the console **28** and can be located proximate to at least the fifth surface **27-5** (e.g., a front surface) of the console **28**, as described herein. Having the docking port **37** be located in a side surface such as the first surface **27-1** of the console **28** and also be located proximate to at least the fifth surface **27-5** of the console can promote aspects herein such as permitting, the controls (e.g., the controls **50** as illustrated in FIG. **5**) and/or the display **52** of the removable controller **36** to be accessible when the removable controller **36** is disposed in the docking port **37**. For instance, FIG. **7** describes an example where both the controls (e.g., the speed controls) and the display **52** are accessible and remain enabled when the removable controller **36** is disposed in or otherwise docked with the console **28** via the docking port **37**, as detailed herein. That is, in some instances, the docking port **37** is configured to expose at least the display **52** of the removable controller **36** when the removable controller **36** is disposed in or otherwise docked with the console **28** via the docking port **37**. In such instances, the removable controller **36** can be configured to continue the display of status indicators (e.g., the second set of status indicators) when the removable controller **36** is removably coupled to the console **28**. In such instances, the removable controller **36** can be configured to continue providing power to and receiving signals from the speed controls (e.g., speed controls **62**, **64** as illustrated in FIG. **5**). For instance, the removable controller **36** may be configured to control the speed of the blood pump at least with the controls **50**. For instance, in some examples, the controls **50** can include speed controls that are configured to control the speed of the blood pump when the removable controller **36** is coupled to and when the removable controller **36** is decoupled from the docking port **37** of the console **28**. In such instances, the console **28** may be without dedicated speed controls therein, which may reduce a size and/or cost of the console **28**. However, in some instances the console **28** may include dedicated speed controls **30**, as illustrated in FIGS. **3-4**, operational to control the speed of the blood pump when the removable controller **36** is docked with the console **28**.

[0083] In some embodiments, the docking port **37** and the removable controller **36** can be configured such that a surface of the docking port **37** is coplanar with an adjacent surface of the removable controller **36** when the removable controller **36** is disposed in the docking port **37**. Alternatively or in addition, the first surface **39-1** of the removable controller **36** can be coplanar with an outer surface of the console **28** when the removable controller **36** is disposed entirely within the docking port **37** of the console **28**. For example, the first surface **39-1** of the removable controller **36** can be coplanar with the first surface **27-1** of the console **28**. Having at least one outer surface of the removable controller **36** be coplanar with at least one outer surface of the console **28** can promote aspects herein such as minimizing a size of the console **28** and/or protecting the removable controller **36**.

[0084] In some embodiments, a portion of the removable controller **36** may be visible and physically accessible when the removable controller **36** is disposed in or otherwise docked with the docking port **37**. For instance, at least one surface of the removable controller **36** may be visible and physically accessible from exterior of the console **28** when the removable controller **36** is disposed in or otherwise docked with the docking port **37**. Having at least one surface of the removable controller **36** be visible and physically accessible can promote aspects herein such as permitting a blood pump (not illustrated in FIGS. **3-4**) to remain coupled to a connector (e.g., the first connector **56**) located on the visible and physically accessible surface of the removable controller **36**.

[0085] FIG. **5** is a schematic diagram showing a plurality of surfaces of the removable controller **36**. As illustrated in FIG. **5**, the plurality of surfaces of the removable controller **36** can include those surfaces previously discussed such as the first surface **39-1**, the second surface **39-2**, the third surface **39-3**, and the fifth surface **39-5**.

[0086] The first surface **39-1** (e.g., the first side surface) can include the first connector **56**. As mentioned, the first connector **56** can be configured to communicatively couple the removable controller **36** with the blood pump via the cable **26**.

[0087] The second surface **39-2** (e.g., the second side surface) can include the second connector **66**.

As mentioned, the second connector **66**, can be configured to communicatively couple the removable controller **36** with the console **28**. For instance, the second connector **66** can be configured to communicatively couple the removable controller **36** to the console **28** when the removable controller **36** is disposed in or otherwise docked with a docking port of the console **28**, as described herein.

[0088] The first connector **56** and/or the second connector **66** may be connectors manifested as ports and/or protrusions. For instance, as illustrated in FIG. **5**, the first connector **56** and the second connector **66** can each be ports. Having both the first connector **56** and the second connector **66** be formed as a respective port may promote aspects herein such as providing a removable controller **36** with generally planar outer surfaces (e.g., without protruding connectors) and thereby minimizing a propensity for the removable controller **36** to snag on or otherwise inadvertently interact with the patient and/or other objects when the removable controller is decoupled from the console **28**. However, the use of other types of connectors is possible.

[0089] The third surface **39-3** (e.g., the top surface) can include controls **50** such as the speed controls **62**, **64** and/or a speed lock **65**. The speed control **62** can increase a speed of a blood pump coupled via the first connector **56** to the removable controller **36**. The speed control **64** can decrease a speed of the blood pump. The speed lock **65** may be configured to lock the blood pump at a given speed and thereby negate any inputs to the speed controls **62**, **64** when the speed lock is in a locked position. The speed lock **65** may be configured to permit alteration of a speed of the blood pump via an input to the speed controls **62**, **64** when the speed lock is in an unlock position. The speed indicator **63** can include a current speed and/or speed level of the blood pump.

[0090] The fifth surface **39-5** (e.g., the front surface) can include a display **52**. The display **52** can be similar in form and function to other displays discussed herein. The display can include visual parameter representations **542***a*, **544***a*, **546***a*, **548***a*, **550***a* which are horizontally aligned with one another along the display **52** along the horizontal axis **550**. FIG. **5** further illustrates that each of the visual parameter representations **542***a*, **544***a*, **546***a*, **548***a*, **550***a* may include a symbol (e.g., icon) which may represent a particular functional or operational parameter of the percutaneous circulatory support systems **10**, **11** or a physiological parameter of the patient **16**.

[0091] As illustrated in FIG. **5**, the first visual representation **542***a* corresponds to a status of an electrical connection between the cable **26** (e.g. the electrical cable) and controller **36**/console **28**, the second visual representation **544***a* corresponds to a status of a motor rotating within the blood pump, the third visual representation **546***a* corresponds to a status of blood flow (e.g., in terms of a pump flow rate or blood flow rate through the blood pump), the fourth visual representation **548***a* corresponds to a status of position of the blood pump within a vasculature of the patient **16**, and the fifth visual representation **550***a* corresponds to values of mean arterial pressure (e.g., 99 mmHg), a blood pressure, an aortic pressure (AO), a left ventricular pressure (LV), and/or other suitable pressure or pressure related values. The parameter values may be updated at any suitable interval including, but not limited to, every beat of the patient's heart, every second, every minute, every five minutes, every thirty minutes, and/or over other suitable periods of time.

[0092] Alternatively or in addition, different parameters associated with the percutaneous circulatory support system **10** and/or the patient **16** may be represented on the display **52**. For instance, indicators of a status (e.g., a current percentage charge) of the battery in the removable controller **36** may be displayed. The visual representations can be any suitable information related to the patient, the removably controller **36**, and/or operation of the blood pump **24**. In some examples, the visual representation may be based on live information related to the patient and/or operation of the blood pump **24**. That is, the removable controller **36** can be configured to provide real-time or near real-time indications of a status of the patient and/or operation of the blood pump **24**.

[0093] Each of the visual parameter representations can have a warning and/or alarm status indicator associated therewith. For instance, as illustrated in FIG. **5**, each of the visual parameter representations can have two respective visual parameter representations associated therewith. For example, FIG. **5** illustrates the first-level warning and/or alarm status indicators **542***b*, **544***b*, **546***b*, **548***b*, **550***b* (e.g., cautionary warning and/or alarm) represented by a circle, whereby each of the circle-shaped status indicators **542***b*, **544***b*, **546***b*, **548***b*, **550***b* are aligned with one another along the horizontal axis **554**. Further, FIG. **5** illustrates the second-level warning and/or alarm status indicators **542***c*, **544***c*, **546***c*, **548***c*, **550***c* (e.g., critical warning and/or alarm) represented by a circle, whereby each of the circle-shaped status indicators **542***c*, **544***c*, **546***c*, **548***c*, **550***c* are aligned with one another along the horizontal axis **556**. FIG. **5** illustrates that the first-level warning and/or alarm status indicators **542***b*, **544***b*, **546***b*, **548***b*, **550***b*, the second-level warning and/or alarm status indicators **542***c*, **544***c*, **546***c*, **548***c*, **550***c* shown on display **52** may be vertically aligned with their corresponding visual parameter representations **542***a*, **544***a*, **546***a*, **548***a*, **550***a*, respectively, along a given vertical axis (e.g., the first-level warning and/or alarm may be located below the corresponding visual parameter representation and the second-level warning and/or alarm may be located above the corresponding visual parameter representation). However, other alignments are possible. As described herein, any of the status indicators (or visual parameter representations) illustrated in FIG. **5** may be represented by any shape, including a square, triangle, rectangle, oval, circle, star, polygon, etc.

[0094] As discussed herein, each of the visual parameter representations **542***a*, **544***a*, **546***a*, **548***a*, **550***a* illustrated and described herein, may identify (e.g., represent, symbolize, etc.) a particular functional or operational parameter of the percutaneous circulatory support system **10** or a physiological parameter the patient **16**, while also conveying information relating to the “status” of that particular parameter during a medical procedure. For example, the display **52** may be configured to present status information relating to a functional parameter of the percutaneous circulatory support systems **10**, **11** or a physiological parameter the patient **16** in a simple, easy-to-understand format such that a clinician my easily determine from the display **52** the status of each parameter represented by the visual parameter representations **542***a*, **544***a*, **546***a*, **548***a*, **550***a*. For instance, the visual parameter representations **542***a*, **544***a*, **546***a*, **548***a*, **550***a* may be illuminated during normal (e.g., routine) operation in which the associated parameter is functioning normally. For example, the first visual representation **542***a* may be illuminated (e.g., white light) when the electrical connection between the cable **26** (e.g. the electrical cable) and controller **36**/console **28** is established, the second visual representation **544***a* may be illuminated (e.g., white light) when the motor is rotating normally, the third visual representation **546***a* may be illuminated (e.g., white light) when normal blood flow is detected through the blood pump, the fourth visual representation **548***a* may be illuminated (e.g., white light) when the system detects that the blood pump is correctly positioned across the aortic valve of the patient, and the fifth visual representation **550***a* may be illuminated (e.g., white light) when the sensed mean arterial pressure, or other a blood pressure, is within a normal range.

[0095] Accordingly, in some instances, the first-level and/or second-level warning and/or alarm status indicators of a given visual parameter representations may provide real-time, stepwise (e.g., progressive), status information (e.g., an alert indication) corresponding to an abnormal functional or a physiological parameter of the percutaneous circulatory support systems **10**, **11** or patient **16**.

[0096] Illumination of one or more of the first-level warning and/or alarm status indicators **542***b*, **544***b*, **546***b*, **548***b*, **550***b* is indicative of a cautionary warning and/or alarm associated with abnormal function of the percutaneous circulatory support systems **10**, **11** and/or abnormal function or out of range physiological parameter of the percutaneous circulatory support systems **10**, **11** or patient **16**. Accordingly, in some instances the visual parameter representation **542***a* shown in FIG. **5** may include a symbol which conveys to a clinician that the visual parameter array **542***a* represents the normal electrical connection between the cable **26** and console **28** of the percutaneous circulatory support system **10**. The display of the first-level warning and/or alarm status indicator **544***b* may convey that the status of the electrical connection between the cable **26** and console **28** is not in normal operating condition (e.g., an insufficient amount of current is being transferred between the console **28** and the circulatory support device **12** along the cable **26** and/or no electrical connection is established). In other words, displaying any of the first-level warning and/or alarm status indicators **542***b*, **544***b*, **546***b*, **548***b*, **550***b* may convey to a clinician that the parameter to which the first status indicator **542***b*, **544***b*, **546***b*, **548***b*, **550***b* relates is not operating normally and intervention is necessary.

[0097] Illumination of one or more of the second-level warning and/or alarm status indicators **542***c*, **544***c*, **546***c*, **548***c*, **550***c* is indicative of a serious or critical warning and/or alarm associated with abnormal function of the percutaneous circulatory support systems **10**, **11** and/or abnormal function or out of range physiological parameter of the percutaneous circulatory support systems **10**, **11** or patient **16**. In some embodiments, a second-level warning and/or alarm status indicator such as the second-level warning and/or alarm status indicator **542***c* of the parameter represented by the visual parameter representation **542***a* (e.g., the electrical connection between the cable **26** and console **28**) can be displayed. The second status indicator **542***c* may be configured to indicate that the visual parameter representation **542***a* has changed to a “serious” condition or “critical” condition which may indicate that the functional, operational or physiological parameter represented by the visual parameter representation **542***a* has moved out of the normal or operating range or a cautionary operating range and into a serious or critical condition, whereby user action maybe be required and the parameter may need improvement, adjustment, etc. (e.g., the electrical current being transferred between the console **28** and the circulatory support device **12** along the cable **26** may be in compromised condition).

[0098] Further, in some examples, the processing system of the percutaneous circulatory support systems **10**, **11** may be programmed to define a range of values, minimum threshold, maximum threshold, etc. that defines the “serious” or “critical” operating condition of the particular parameter to which the given second-level warning and/or alarm status indicators **542***c*, **544***c*, **546***c*, **548***c*, **550***c* relate.

[0099] FIG. **6** is a schematic diagram showing a removable controller **36** coupled to a patient **16**. As illustrated in FIG. **6**, the removable controller **36** can be coupled to a wearable device **78** that can be worn externally by the patient **16**. The wearable device **78** may be a belt, a harness, sling, or other type of apparatus that is configured to be worn externally by the patent **16** and receive the removable controller **36** during operation of the blood pump away from the console **28**. For instance, the wearable device **78** may include a pouch or other cavity in which the removable controller **36** can be disposed and maintain a coupling with the blood pump when disposed therein. Alternatively or in addition, the wearable device **78** may include a mechanical coupling device such as flange or other type of mechanical component that is configured to interface (e.g., friction fit) with a corresponding mechanical component (e.g., a third connector) on the removable controller **36**. For instance, the removable controller **36** can include a third connector (not illustrated) that is configured to be mechanically coupled the removable controller **36** to the wearable device **78** which is manifested as harness, as illustrated in FIG. **6**. The removable controller **36** may be configured to be mechanically coupled to the wearable device **78** while also being coupled via the cable **26** to a blood pump inside the patient **16**. Having the removable controller **36** be decoupled from a console and be coupled to the wearable device **78** can promote aspects herein such as permitting live control and/or monitoring (e.g., via the display **52**) of a blood pump disposed inside the patient **16** and yet permitting ease of transport of the patient **16**. For instance, the removable controller **36** can be communicatively coupled to the blood pump via a cable **86** when the removable controller is coupled to and when the removable controller **36** is decoupled from a console, as described herein. Cable **86** may be analogous to or similar to cable **26** as described herein.

[0100] FIG. **7** is a schematic diagram of a portion of another illustrative circulatory support system **772** with a removable controller **36** coupled to or otherwise docked with a console **28**. FIG. **7** is similar to FIG. **3**, with the change that the docking port **37** (which extends into the first surface **27-1**) is located adjacent to the third surface **27-3** (e.g., the top surface) and the fifth surface **27-5** (e.g., the front surface) of the console **28**. There may be one or more windows (e.g., passages) extending through a portion of a surface of the console **28** to expose a portion of the removable controller **36** disposed in the console **28**. For instance, the third surface **27-3** (top surface) of the console **28** can include a window **29** to expose the controls **50** of the removable controller **36** when the removable controller **36** is disposed in the docking port **37**. Similarly, the fifth surface **27-5** (front surface) of the console **28** can include a window to expose the display **52**. Thus, in some instances the controls **50** and the display **52** may be accessible when the removable controller **36** is disposed in the docking port **37**.

[0101] In such instances, the controls **50** and the display **52** of the removable controller **36** can be configured to remain active when the removable controller **36** is disposed in the docking port **37**. That is, the controls **50** may permit control of the pump speed, etc. and the display **52** may continue to display information associated with the blood pump and/or the patient when the removable controller **36** is disposed in the docking port **37**. In some instances, the display **52** and/or the controls **50** may provide a degree of redundancy and thereby be available in the event of the same display and/or same controls in the console **28**. However, in some instances the display **52** and/or the controls **50** may take the place of a display (e.g., display **32**) and/or controls (e.g., controls **30**) that typically would be employed in the console **28**. For instance, the controls **50** may include the only speed controls in the percutaneous circulatory support system **10**, in some instances. Having the display **52** and/or the controls **50** take the place of a display (e.g., display **32**) and/or controls (e.g., controls **30**) that typically would be employed in the console **28** may reduce a cost and/or size of the console **28**.

[0102] In some instances, the display **52** may be configured to display at least a first set of status indicators corresponding to parameters related to operation of the blood pump. In such instances, another display in the console **28** such as the display **32** and/or the display **52**, as illustrated in FIG. **4**) may be configured to display a second set of status indicators corresponding to parameters related to operation of the blood pump. In some embodiments, the second set of status indicators can be the same as the first set of status indicators. For instance, the display **32** may be configured to a second set of status indicators corresponding to parameters related to operation of the blood pump that are the same (e.g., are each identical to) the first set of status indicators corresponding to the parameters related to operation of the blood pump display by the display **52**.

[0103] However, in some instances, the display **52** can be configured to display at least a first set of status indicators corresponding to parameters related to operation of the blood pump that is different than a second set of status indicators corresponding to parameters related to operation of the blood pump displayed by another display. For instance, the display **52** can be configured to display at least a first set of status indicators corresponding to parameters related to operation of the blood pump that is different than a second set of status indicators corresponding to parameters related to operation of the blood pump displayed by the display **52**.

[0104] The display **52** may have a first surface area with limited dimensions for displaying screens due to the portable nature of removable controller **36**. The display **52** can have a surface area that is less than a surface area of a display in the console such as the display **54**. The larger surface area of the console display **54** relative to the surface area of the display **52** of the removable controller **36** may allow for additional information and/or data to be provided or accessible to the user relative to the information and/or data provided on screens of the console display **54** and/or facilitate other aspects herein such as promoting ease of transport of the removable controller **36**

[0105] To facilitate the surface area with limited dimensions and/or for other purposes, screens configured for display on the display **52** may have a preconfigured, fixed layout, may be limited in the data and/or information provided (e.g., displayed), and may provide the limited data and/or information in a clear manner that facilitates addressing current conditions of the patient and/or the blood pump **24**. Example screens for display on the display **52** are described herein, but other suitable configurations of screens for display on the display **52** are contemplated.

[0106] In some configurations, the removable controller **36** can be configured to communicate with the console **28** directly through a wired or wireless connection. Additionally or alternatively, the removable controller **36** (e.g., a computing device of the removable controller **36** or a computing device encompassing the removable controller **36**) may be configured to communicate with the console **28** through a server, a cloud computing system, and/or other suitable computing device(s).

[0107] In some instances, the console **28** may be configured to store data (e.g., live data, historical data, etc.) related to operation of the blood pump **24** (e.g., patient and/or blood pump data) and/or send the data related to operation of the blood pump **24** to a server (e.g., a server that is part of a LAN and/or a server that is part of a WAN). In some examples, the console **28** and/or the server may include non-volatile memory to facilitate storing data at the console **28** and/or the server.

[0108] It should be understood that this disclosure is, in many respects, only illustrative. Changes may be made in details, particularly in matters of shape, size, and arrangement of steps without exceeding the scope of the disclosure. This may include, to the extent that it is appropriate, the use of any of the features of one example embodiment being used in other embodiments. The scope of the disclosure is, of course, defined in the language in which the appended claims are expressed.

Claims

1. A circulatory support system comprising: a blood pump; and a removable controller configured to control operation of the blood pump and be removably coupled to a console, the removable controller comprising: a display configured to display status indicators corresponding to parameters related to operation of the blood pump; speed controls configured to control a speed of the blood pump; a first connector configured to communicatively couple the removable controller with the blood pump; and a second connector configured to communicatively couple the removable controller with the console.

2. The system of claim 1, wherein: the first connector is disposed at a first surface of the removable controller; and the second connector is disposed at a second surface of the removable controller.

3. The system of claim 2, wherein the first surface is a first side surface and the second surface is a second side surface that is opposite the first side surface.

4. The system of claim 1, wherein the removable controller is configured to be disposed at least partially within a docking port of the console.

5. The system of claim 4, wherein the removable controller is configured to be disposed entirely within the docking port of the console.

6. The system of claim 5, wherein the first surface is coplanar with an outer surface of the console when the removable controller is disposed entirely within the docking port of the console.

7. The system of claim 1, wherein the removable controller is configured such that at least the first connector remains physically accessible when the removable controller is removably coupled to the console.

8. The system of claim 1, wherein at least the first connector and the display remain accessible when the removable controller is removably coupled to the console.

9. The system of claim 1, wherein the removable controller is configured to be in wired communication, wireless communication, or both, with the console.

10. The system of claim 1, wherein the removable controller includes a rechargeable battery.

11. The system of claim 10, wherein the removable controller is configured to receive wireless power, wired power, or both, to recharge the rechargeable battery.

12. The system of claim 1, including a wearable device configured to receive the removable controller, and wherein the removable controller is configured to be removably coupled to the wearable device.

13. The system of claim 12, wherein the removable controller includes a third connector disposed on an outer surface of the removable controller, and wherein the third connector is configured to removably couple the removable controller to the wearable device.

14. The system of claim 1, wherein the removable controller is configured to remain communicatively coupled to the blood pump when the removable controller is removably coupled to the console to operate the blood pump.

15. The system of claim 14, wherein the removable controller is configured to remain communicatively coupled to the blood pump when the removable controller is decoupled from the console to operate the blood pump.

16. A circulatory support system comprising: a blood pump; a console including: a display configured to display at least a first set of status indicators corresponding to parameters related to operation of the blood pump; and a docking port; and a connector; and a removable controller comprising: a user interface configured to display a second set of status indicators corresponding to parameters related to operation of the blood pump; speed controls configured to control a speed of the blood pump; a first connector configured to communicatively couple the removable controller with the blood pump; and a second connector configured to connect to the connector of the console when the removable controller is docked with the console via the docking port to communicatively couple the removable controller to the console.

17. The system of claim 16, wherein: the docking port is configured to obscure at least the display of the removable controller when the removable controller docked with the console via the docking port; and the removable controller is configured to cease display of the second set of status indicators when communicatively coupled to the console.

18. The system of claim 16, wherein: the display of the removable controller remains exposed when the removable controller is docked with the console via the docking port; and the removable controller is configured to continue to display the second set of status indicators when communicatively coupled to the console.

19. A circulatory support system comprising: a blood pump; a console including: a display configured to display at least a first set of status indicators corresponding to parameters related to operation of the blood pump; and a docking port; a connector; and a first battery; and a removable controller comprising: a display configured to display a second set of status indicators corresponding to parameters related to operation of the blood pump; speed controls configured to control a speed of the blood pump; a first connector configured to communicatively couple the removable controller with the blood pump; a second connector configured to connect to the connector of the console when the removable controller is docked with the console via the docking port to communicatively couple the removable controller with the console; and a second battery.

20. The system of claim 19, wherein: the removable controller is configured to maintain operation of the blood pump when undocked from the console.